

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (5 points) Find Laplace transforms of the following functions:

(a) $f(t) = -2e^{3t}$

$$\mathcal{L}\{-2e^{3t}\} = -2\mathcal{L}\{e^{3t}\} = -2 \cdot \frac{1}{s-3} = -\frac{2}{s-3} \quad (\text{for } s > 3)$$

(b) $f(t) = 2\sin 3t + \cos 3t$

$$\begin{aligned} \mathcal{L}\{2\sin 3t + \cos 3t\} &= 2\mathcal{L}\{\sin 3t\} + \mathcal{L}\{\cos 3t\} \\ &= 2 \cdot \frac{3}{s^2+9} + \frac{s}{s^2+9} \\ &= \frac{s+6}{s^2+9} \quad (\text{for } s > 0) \end{aligned}$$

Problem 2. (5 points) Find inverse Laplace transforms of the following functions:

(a) $F(s) = \frac{6}{s^4}$

$$\mathcal{L}^{-1}\left\{\frac{6}{s^4}\right\} = t^3$$

(b) $F(s) = \frac{8}{s^2-4}$

$$\mathcal{L}^{-1}\left\{\frac{8}{s^2-4}\right\} = 4\mathcal{L}^{-1}\left\{\frac{2}{s^2-2^2}\right\} = 4\sinh 2t$$